

Nikolaos Bellas

Professor

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Date: Spring 2026

RESEARCH INTERESTS

Architecture of computing systems: ML for system design; reconfigurable systems; heterogeneous systems; embedded systems; approximate computing; low-power design; computer architecture.

Software: EDA tools for architectural synthesis; system software for reliable and approximate computing; multimedia.

EMPLOYMENT HISTORY

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| 12.2019 – now | Professor. ECE Department, University of Thessaly, Volos, Greece. |
| 04.2008 – 12.2019 | Associate Professor. ECE Department, University of Thessaly, Volos, Greece. |
| 09.2007 – 05.2008 | Industrial Consultant. Ported the AVS video decoding standard to the Tensilica Diamond processor using Tensilica instruction extensions and dual core API. |
| 10.2001 – 03.2007 | Principal Staff Engineer. Embedded Systems Research, Motorola Inc., Schaumburg, IL. <ul style="list-style-type: none">• Technical lead of a five-person team working on reconfigurable computing.• Designed CAD tool that compiles streaming applications written in a high level language to accelerators in a System On FPGA platform.• Chief architect of two ASIC chips that perform image processing for wireless multimedia applications for the next generation camera-enabled Motorola phones.• Participated in the design of a System On Chip ASIC design for an automotive application based on the ARM9 RISC processor. |
| 05.1999 – 10.2001 | Senior Staff Engineer. Multimedia Architecture Lab, Motorola Inc., Schaumburg, IL. <ul style="list-style-type: none">• Architect of a scalable, programmable architecture for MPEG4 video and JPEG image compression.• Designed a programmable Motion Estimation module and defined an Instruction Set Architecture. |

- Designed the image-processing module that interfaces to the image sensor.
- Participated in chip hardware testing and image processing algorithm design for Motorolas first 3G camera cell phone.

12.1998 – 05.1999 — **Postdoctoral Research Associate.** ECE Department, University of Illinois at Urbana-Champaign (UIUC).

05.1996 – 08.1996 — **Research Intern.** Design Technology Labs, Intel Corp., Santa Clara, CA.

05.1995 – 08.1995 — **Research Intern.** Silicon Graphics (SGI), Mountain View, CA.

08.1993 – 12.1998 — **Graduate Research Assistant.** ECE Department, UIUC.

EDUCATION

PhD in Electrical and Computer Engineering, University of Illinois at Urbana-Champaign (1998).

Dissertation Title: “Architectural and Compiler Support for Energy Reduction in High Performance Microprocessors”.

Advisors: Ibrahim Hajj and Constantine Polychronopoulos.

MSc in Electrical and Computer Engineering, University of Illinois at Urbana-Champaign (1995).

Thesis Title: “A Novel Design for Testability Technique Using State Space Information”.

Advisor: Daniel Saab.

Diploma in Computer Engineering and Informatics, University of Patras, Greece (1992).

CHAPTERS IN BOOKS

- [B1] Panos Koutsovasilis, Christos Kalogirou, Konstantinos Parasyris, Christos D. Antonopoulos, Nikolaos Bellas, Spyros Lalas. “Exploiting Reduced Voltage Margins: From the Node to the Datacentre Level”. In *Computing at the EDGE: New Challenges for Service Provision*, Georgios Karakonstantis, Charles J. Gillan (editors). Springer, 2022, ISBN 3-030-74535-X.

JOURNAL ARTICLES

- [J15] Foivos Pournaropoulos, Alexandros Patras, Christos D. Antonopoulos, Nikos Bellas, Spyros Lalas. Fluidity: Providing Flexible Deployment and Adaptation Policy Experimentation for Serverless and Distributed Applications spanning Cloud-Edge-Mobile Environments. *Future Generation Computer Systems*. March 2024.
- [J14] Maria-Rafaela Gkeka, Alexandros Patras, Nikolaos Tavoularis, Stylianos Piperakis, Emmanouil Hourdakos, Panos Trahanias, Christos D. Antonopoulos, Spyros Lalas, Nikolaos Bellas. Reconfigurable System-on-Chip Architectures for Robust Visual SLAM on Humanoid Robots. *ACM Transactions on Embedded Computing Systems*. Vol. 23, No.2,

March-April 2023.

- [J13] P. Koutsovasilis, C. Antonopoulos, N. Bellas, S. Lalis, G. Papadimitriou, A. Chatzidimitriou, and D. Gizopoulos. The Impact of CPU Voltage Margins on Power-Constrained Execution. *IEEE Transactions on Sustainable Computing*, vol. 7, no. 1, pp. 221-234, Jan.-March 2022.
- [J12] Panos Koutsovasilis, Konstantinos Parasyris, Christos D. Antonopoulos, Nikolaos Bellas, Spyros Lalis. Dynamic Undervolting to Improve Energy Efficiency on Multicore x86 CPUs. *IEEE Transactions on Parallel and Distributed Systems (TPDS)*. pg. 2851-2864. Vol. 31 (12). December 2020.
- [J11] Konstantinos Parasyris, Vassilis Vassiliadis, Christos D. Antonopoulos, Spyros Lalis, Nikolaos Bellas. Significance-Aware Program Execution on Unreliable Hardware. *ACM Transactions on Architecture and Code Optimization (TACO)*. Vol. 14 (2). April 2017.
- [J10] Vassilis Vassiliadis, Charalampos Chaliou, Konstantinos Parasyris, Christos D. Antonopoulos, Spyros Lalis, Nikolaos Bellas, Hans Vandierendonck, Dimitrios S. Nikolopoulos. Exploiting Significance of Computations for Energy-Constrained Approximate Computing. *International Journal of Parallel Programming (IJPP)*. pg. 1078-1098. March 2016.
- [J9] K. Krommydas, Wu-chun Feng, Christos D. Antonopoulos, and N. Bellas. OpenDwarfs: Characterization of Dwarf-Based Benchmarks on Fixed and Reconfigurable Architectures. *Journal of Signal Processing Systems*. US Springer, 85(3), pg. 373-392. 2016.
- [J8] M. Owaida, G. Falcao, J. Andrade, C. Antonopoulos, N. Bellas, M. Purnaprajna, D. Novo, G. Karakonstantis, A. Burg, and P. Ienne. Enhancing design space exploration by extending CPU/GPU specifications onto FPGAs. *ACM Transactions on Embedded Computing Systems (TECS)*. Vol 14(2). March 2015.
- [J7] Dimitrios Nikolopoulos, Hans Vandierendock, Nikolaos Bellas, Christos D. Antonopoulos, Spyros Lalis, Georgios Karakonstantis, Andreas Burg, Uwe Naumann. Energy Efficiency through Significance-Based Computing. *IEEE Computer*. Vol. 47. Issue 7. Pg. 82-85. July 2014.
- [J6] Maria Koziri, Dimitris Zacharis, Ioannis Katsavounidis, Nikolaos Bellas. Implementation of the AVS Video Decoder on a Heterogeneous Dual-Core SIMD Processor. *IEEE Transactions on Consumer Electronics*, vol. 57, No. 2, pp. 673-681, May 2011.
- [J5] Seda Ogrenci Memik, Nikolaos Bellas, Somsubhra Mondal. Pre-synthesis Area Estimation of Reconfigurable Streaming Accelerators. *IEEE Transactions on Computer-Aided Design*, Volume: 27, No: 11, pp. 2027-2038, November 2008.
- [J4] Nikolaos Bellas, Sek Chai, Malcolm Dwyer, Dan Linzmeier. Mapping streaming architectures on reconfigurable platforms. *ACM SIGARCH Computer Architecture News*. Volume 35, Issue 3, Pages: 2 – 8, June 2007.
- [J3] Nikolaos Bellas, Ibrahim Hajj, Constantine Polychronopoulos. Using dynamic cache management techniques to reduce energy in general-purpose Processors. *IEEE Transactions on VLSI Systems*, Volume: 8, Issue: 6, pp. 693-708, December 2000.
- [J2] Nikolaos Bellas, Ibrahim Hajj, Constantine Polychronopoulos, George Stamoulis. Architectural and Compiler Techniques for Energy Reduction in High-Performance Microprocessors. *IEEE Transactions on VLSI Systems*, Special Issue on Low Power, Volume:8, Issue:3, pp. 317-326, June 2000.

- [J1] Amber-Roy Chowdhury, Nikolaos Bellas, Prithviraj Banerjee. Algorithm-Based Error Detection Schemes for Iterative Solution of Partial Differential Equations. *IEEE Transactions on Computers*, pp. 394-407, Vol. 45, Number 4, April 1996.

CONFERENCE AND WORKSHOP ARTICLES

- [C60] Bowen Sun, Christos Antonopoulos, Evgenia Smirni, Bin Ren, Nikolaos Bellas, Spyros Lalīs. Green or Fast? Learning to Balance Cold Starts and Idle Carbon in Serverless Computing. *26th IEEE International Symposium on Cluster, Cloud, and Internet Computing (CCGrid)*. May 18-21, 2026. Sydney, Australia.
- [C59] Alexandros Patras, Christos D. Antonopoulos, Spyros Lalīs, Nikolaos Bellas. DPUConfig: Optimizing ML Inference in FPGAs Using Reinforcement Learning. *Design, Automation & Test in Europe Conference & Exhibition (DATE)*. April 20-22, 2026. Verona, Italy.
- [C58] Jiexiong Guan, Zhenqing Hu, Christos D. Antonopoulos, Nikolaos Bellas, Spyros Lalīs, Evgenia Smirni, Gang Zhou, Gagan Agrawal, Bin Ren. Modeling Texture Memory and Mobile GPU Performance to Accelerate DNN Computations. *39th International Conference on Supercomputing (ICS)*. June 8-11, 2025. Salt Lake City, UT.
- [C57] Ioanna-Maria Panagou, Nikolaos Bellas, Lorenzo Moneta and Sanjiban Sengupta. Accelerating Machine Learning Inference on GPUs with SYCL. *12th International Workshop on OpenCL and SYCL (IWOCCL)*. April 8-11, 2024. Chicago, IL.
- [C56] Alexandros Patras et al. A Minimal Testbed for Experimenting with Flexible Resource and Application Management in Heterogeneous Edge-Cloud Systems. *International Conference on Embedded Wireless Systems and Networks (EWSN)*. September 25-27, 2023. Rende, Italy.
- [C55] Antonios-Kyriillos Chatzimichail, Charalampos Antoniadis, Nikolaos Bellas, Yehia Mas-soud. Low Power Hardware Architecture for Sampling-Free Bayesian Neural Networks Inference. *IEEE International Symposium on Circuits and Systems (ISCAS)*. May 21-25, 2023. Monterrey, CA. (Poster).
- [C54] Ioanna-Maria Panagou et al. FPGA Roofline modeling and its Application to Visual SLAM. *32th International Symposium on Field Programmable Logic and Applications (FPL)*. August 29-September 2, 2022. (Short paper).
- [C53] Christos Kalogirou, Christos D. Antonopoulos, Spyros Lalīs, and Nikolaos Bellas. Dynamic Management of CPU Resources Towards Energy Efficient and Profitable Datacentre Operation. *25th Workshop on Job Scheduling Strategies for Parallel Processing (JSSPP), in conjunction with IPDPS 2022*. June 3, 2022, Lyon, France.
- [C52] Maria Rafaela Gkeka et al. FPGA Accelerators for Robust Visual SLAM on Humanoid Robots. *30th ACM/SIGDA International Symposium on Field-Programmable Gate Arrays (FPGA)*. Feb. 27- March 1, 2022. Virtual Event. (Poster).
- [C51] Nikolaos Bellas, et al. Architectures for SLAM and Augmented Reality Computing. *31th International Symposium on Field Programmable Logic and Applications (FPL)*. August 30-September 3, 2021. Virtual event. (Project paper).
- [C50] Emmanouil Maroudas, Christos Antonopoulos, Nikolaos Bellas and Spyros Lalīs. Exploring the Potential of Context-Aware Dynamic CPU Undervolting. *ACM International Conference on Computing Frontiers*. May 11-13, 2021. Virtual Conference. (Best Paper Award).

- [C49] Maria-Rafaela Gkeka et al. FPGA Architectures for Approximate Dense SLAM Computing. *Design, Automation & Test in Europe Conference & Exhibition (DATE)*. February 1-5, 2021. Virtual Conference (Nominated for Best Paper Award).
- [C48] Christos Kalogirou et al. Increasing the Profit of Cloud Providers through DRAM Operation at Reduced Margins. *20th IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGrid)*. May 11-14, 2020. Melbourne, Australia.
- [C47] Christos Kalogirou et al. Exploiting CPU Voltage Margins to Increase the Profit of Cloud Infrastructure Providers. *19th IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGrid)*. May 14-17, 2019. Larnaca, Cyprus.
- [C46] Maria Rafaela Gkeka, Nikolaos Bellas, Christos D. Antonopoulos. Comparative Performance Analysis of Vulkan Implementations of Computational Applications. *7th International Workshop on OpenCL (IWOCCL)*. May 13-15, 2019. Boston, MA.
- [C45] K. Parasyris et al. Exploring the Effects of Code Optimizations on CPU Frequency Margins. *Workshop in Approximate and Transprecision Computing on Emerging Technologies (ATCET), in conjunction with the International Supercomputing Conference (ISC)*. June 28th, 2018. Frankfurt, Germany.
- [C44] K. Parasyris et al. A Framework for Evaluating Software on Reduced Margins Hardware. *International Conference on Dependable Systems and Networks (DSN)*. June 25-28, 2018. Luxemburg.
- [C43] Georgios Karakonstantis et al. An Energy-Efficient and Error-Resilient Server Ecosystem Exceeding Conservative Scaling Limits. *IEEE Conference on Design, Automation and Test in Europe (DATE)*. March 2018. Dresden, Germany.
- [C42] Christos Kalogirou et al. Edge and Cloud Provider Cost Minimization by Exploiting Extended Voltage and Frequency Margins. *International Conference on Parallel Computing (PARCO)*. September 12-15, 2017, Bologna, Italy.
- [C41] Ioannis Parnassos et al. Programming Model and Runtime System for Approximation-Aware Heterogeneous Computing. *27th International Symposium on Field Programmable Logic and Applications (FPL)*. September 4-8, 2017. Ghent, Belgium.
- [C40] Ioannis Parnassos et al. SoCLog: A Real-Time, Automatically Generated Logging and Profiling Mechanism for FPGA-based Systems On Chip. *26th International Symposium on Field Programmable Logic and Applications (FPL)*. August 28 – September 2, 2016. Lausanne, Switzerland.
- [C39] Vassilis Vassiliadis et al. Towards Automatic Significance Analysis for Approximate Computing. *International Symposium on Code Generation and Optimization (CGO)*. March 14-16, 2016. Barcelona, Spain.
- [C38] Michalis Spyrou et al. Energy Minimization on Heterogeneous Systems through Approximate Computing. *International Conference on Parallel Computing (PARCO)*. September 1-4, 2015. Edinburgh, UK.
- [C37] Vassilis Vassiliadis et al. A Significance-Driven Programming Framework for Energy-Constrained Approximate Computing. *ACM International Conference on Computing Frontiers*. May 18-21, 2015. Ischia, Italy.
- [C36] Konstantinos Parasyris et al. A Significance-Aware Software Stack for Computing on Unreliable Hardware. *Second Workshop on Approximate Computing Across the System Stack (WACAS)*. March 15, 2015. Istanbul, Turkey.

- [C35] Vassilis Vassiliadis et al. A programming model and runtime system for significance-aware energy-efficient computing. *ACM SIGPLAN 20th Symposium on Principles and Practice of Parallel Programming (PPoPP)*. February 9-11, 2015. San Francisco, CA.
- [C34] Konstantinos Parasyris et al. GemFI: A Fault Injection Tool for Studying the Behavior of Applications on Unreliable Substrates. *International Conference on Dependable Systems and Networks (DSN)*. June 23-26, 2014. Atlanta, GA.
- [C33] Konstantinos Krommydas et al. On the Portability of OpenCL Dwarfs on Fixed and Reconfigurable Parallel Platforms. *25th IEEE International Conference on Application-specific Systems, Architectures and Processors (ASAP)*. June 18-20, 2014, Zurich, Switzerland. (Nominated for Best Paper Award).
- [C32] Muhsen Owaida, Christos D. Antonopoulos, Nikolaos Bellas. A Grammar Induction Method for Clustering of Operations in Complex FPGA Designs. *IEEE Symposium on Field-Programmable Custom Computing Machines (FCCM)*. May 11-13, 2014. Boston, MA
- [C31] Konstantinos Krommydas, Muhsen Owaida, Christos D. Antonopoulos, Nikolaos Bellas, Wu-chun Feng. On the Portability of the OpenCL Dwarfs on Fixed and Reconfigurable Parallel Platforms. *19th IEEE International Conference on Parallel and Distributed Systems (ICPADS)*, December 15-18, 2013, Seoul, Korea.
- [C30] Gabriel Falcao et al. Shortening design time through multiplatform simulations with a portable OpenCL golden model: the LDPC decoder case. *IEEE Symposium on Field-Programmable Custom Computing Machines (FCCM)*. April-May 2012. Toronto, ON.
- [C29] Muhsen Owaida et al. Massively Parallel Programing Models Used as Hardware Description Languages: The OpenCL Case. *International Conference on Computer-Aided Design (ICCAD)*. November 6-10, 2011, San Jose, CA.
- [C28] Konstantinos Krommydas et al. AVS Video Decoder on Multicore Systems: Optimizations and Tradeoffs. *2011 International Conference on Multimedia and Expo (ICME)*. July 2011, Barcelona, Spain.
- [C27] Georgios Karakonstantis et al. Significance Driven Computation on Next Generation Unreliable Platforms. *Design Automation Conference (DAC), Wild And Crazy Ideas Session (WACI)*. June 5-10, 2011, San Diego, CA.
- [C26] Muhsen Owaida et al. Synthesis of Platform Architectures from OpenCL Programs. *IEEE Symposium on Field-Programmable Custom Computing Machines (FCCM)*. May 1-3, 2011, Salt Lake City, UT.
- [C25] Kostas Theocharoulis et al. Implementation and Performance Analysis of Seal Encryption on FPGA, GPU, and Multicore Processors. *IEEE Symposium on Field-Programmable Custom Computing Machines (FCCM)*. May 1-3, 2011.
- [C24] Muhsen Owaida et al. Implementation and Performance Comparison of the Motion Compensation Kernel of the AVS Video Decoder on FPGA, GPU and Multicore Processors. *IEEE Symposium on Field-Programmable Custom Computing Machines (FCCM)*. May 1-3, 2011.
- [C23] Konstantis Daloukas, C.D. Antonopoulos, Nikolaos Bellas. GLOpenCL: OpenCL Support on Hardware- and Software-Managed Cache Multicores. *6th International Conference on High Performance Embedded Architectures & Compilers (HiPEAC)*. January 24-26, 2011, Heraklion, Greece.

- [C22] Konstantinos Krommidas et al. Mapping and Optimization of the AVS Video Decoder on a High Performance Chip Multiprocessor. *International Conference on Multimedia and Expo (ICME)*. July 19-23, 2010, Singapore.
- [C21] Konstantis Daloukas et al. Fisheye Lens Distortion Correction on Multicore and Hardware Accelerator Platforms. *24th International Parallel and Distributed Processing Symposium (IPDPS)*. April 19-23, 2010, Atlanta, GA.
- [C20] Nikolaos Bellas et al. Proteus: An Architectural Synthesis Tool based on the Streaming Programming Model. *19th International Conference on Field Programmable Logic and Applications (FPL)*. August/September 2009, Prague.
- [C19] Sek M. Chai, Nikolaos Bellas, Abelardo Lopez Lagunas. Extending a Stream Programming Paradigm to Hardware Accelerator Platforms. *Symposium on Application Accelerators for High Performance Computing (SAAHPC)*. July 27-31 2009, Champaign, IL.
- [C18] Nikolaos Bellas et al. Mapping the AVS Video Decoder on a Heterogeneous Dual-Core SIMD Processor. *46th Design Automation Conference (DAC)*. July 26-31, 2009, San Francisco, CA.
- [C17] Konstantis Daloukas, C.D. Antonopoulos, Nikolaos Bellas. Implementation of a Wide-angle Lens Distortion Correction Algorithm on the Cell Broadband Engine. *23rd International Conference on Supercomputing (ICS)*. June 8-12, 2009, New York Metro Area, NY.
- [C16] Nikolaos Bellas et al. Real-Time Fisheye Lens Distortion Correction Using Automatically Generated Streaming Accelerators. *IEEE Symposium on Field-Programmable Custom Computing Machines (FCCM)*. April 5-7, 2009, Napa Valley, CA.
- [C15] Nikolaos Bellas et al. An Architectural Framework for Automated Streaming Kernel Selection. *14th Reconfigurable Architectures Workshop (RAW)*. March 2007, Long Beach, CA.
- [C14] Nikolaos Bellas et al. Mapping Streaming Architectures on Reconfigurable Platforms. *Reconfigurable and Adaptive Architectures Workshop (RAAW)*. December 10th, 2006, Orlando, FL.
- [C13] Somsubhra Mondal, Seda O. Memik, Nikolaos Bellas. Pre-synthesis area estimation of reconfigurable streaming accelerators. *16th International Conference on Field Programmable Logic and Applications (FPL)*. August 28-30 2006, Madrid, Spain.
- [C12] Nikolaos Bellas, Arnold Yanof. An Image Processing Pipeline with Digital Compensation of Low Cost Optics for Mobile Telephony. *International Conference on Multimedia and Expo (ICME)*. July 9-12, 2006, Toronto, Canada.
- [C11] Chai, Nikolaos Bellas et al. Reconfigurable Streaming Architectures for Embedded Smart Camera Applications. *2nd IEEE Workshop on Embedded Computer Vision, in conjunction with CVPR*. June 18, 2006, New York, NY.
- [C10] Nikolaos Bellas et al. FPGA implementation of a license plate recognition SoC using automatically generated streaming accelerators. *13th Reconfigurable Architectures Workshop (RAW)*. 25-26 April 2006, Rhodes, Greece.
- [C9] Nikolaos Bellas et al. Template-based generation of streaming accelerators from a high level representation. *International Symposium on Field-Programmable Custom Computing Machines (FCCM)*. April 24-26, 2006, Napa Valley, CA.

- [C8] Somsubhra Mondal, Seda O. Memik, Nikolaos Bellas. Pre-synthesis Queue Size Estimation of Streaming Data Flow Graphs. *International Symposium on Field-Programmable Custom Computing Machines (FCCM)*. April 24-26, 2006, Napa Valley, CA.
- [C7] Sek Chai et al. Stream Memory Subsystem in Reconfigurable Platforms. *2nd Workshop on Architecture Research using FPGA Platforms (WARFP)*. February 12, 2006, Austin, TX.
- [C6] Nikolaos Bellas, Malcolm Dwyer. A programmable, high performance Vector array unit used for Real-time Motion Estimation. *Proceedings of the International Conference on Multimedia and Expo (ICME)*. July 2003, Baltimore, MD.
- [C5] Nikolaos Bellas et al. Energy and Performance Improvements in Microprocessor Design using a Loop Cache. *Proceedings of the International Symposium on Computer Design (ICCD)*. October 1999, Austin, TX.
- [C4] Nikolaos Bellas et al. Using dynamic cache management techniques to reduce energy in a high-performance microprocessor. *International Symposium of Low Power Electronics and Design (ISLPED)*. August 1999, San Diego, CA.
- [C3] Nikolaos Bellas et al. A detailed, transistor-level energy model for SRAM-based caches. *International Symposium of Circuits and Systems (ISCAS)*. June 1999, Orlando, FL.
- [C2] Nikolaos Bellas et al. Architectural and Compiler Support for Energy Reduction in the Memory Hierarchy of High Performance Microprocessors. *Proceedings of the International Symposium of Low Power Electronics and Design (ISLPED)*. August 1998, Monterey, CA.
- [C1] Nikolaos Bellas et al. A new scheme for I-Cache energy reduction in High Performance Processors. *Power-Driven Microarchitecture Workshop, International Symposium On Computer Architecture (ISCA)*. June 1998, Barcelona, Spain.

PATENTS AND PATENT APPLICATIONS

- [P13] US Patent 8,855,441. Sek M. Chai, Malcolm Dwyer, Dan Linzmeier, Ruei-Sung Lin, Nikolas Bellas. Method and apparatus for transforming a non-linear lens-distorted image, October 2014, Motorola Corp.
- [P12] US Patent 8,326,077. Sek M. Chai, Malcolm Dwyer, Dan Linzmeier, Ruei-Sung Lin, Nikos Bellas. Method and apparatus for transforming a non-linear lens-distorted image, December 2012, Motorola Corp.
- [P11] US Patent 7,802,005. Sek M. Chai, Nikos Bellas, Malcolm Dwyer, Dan Linzmeier. Method and apparatus for configuring buffers for streaming data transfer. September 2010, Motorola Corp.
- [P10] US Patent 7,683,948. Arnold Yanof, Nikos Bellas. "System and method for bad pixel replacement in image processing". March 2010, Freescale Corp.
- [P9] US Patent 7,603,492. Sek M. Chai, Nikos Bellas, Malcolm Dwyer, Erica Lau, Zhiyuan Li, Dan Linzmeier. Automatic generation of streaming data interface circuit, October 2009, Motorola Corp.
- [P8] US Patent 7,580,070. Arnold Yanof, Nikos Bellas. "System and method for roll-off correction in image processing", August 2009, Freescale Corp.

- [P7] US Patent 7,441,224. Nikos Bellas, Sek M. Chai, Dan Linzmeier. Streaming kernel selection for reconfigurable processor, October 2008, Motorola Corp.
- [P6] US Patent 7,305,649. Nikos Bellas, Sek Chai, Erica Lau, Zhiyuan Li, Dan Linzmeier. "Automatic generation of streaming processor circuit", December 2007, Motorola Corp.
- [P5] US Patent 7,073,041. Malcolm Dwyer, Nikolaos Bellas. "Virtual Memory Translation Unit for Media Acceleration", July 2006, Motorola Corp.
- [P4] US Patent 6,868,123. Nikolaos Bellas, Malcolm Dwyer. "A programmable, high performance Vector array unit used for Real-time Motion Estimation", March 2005, Motorola Corp.
- [P3] US Patent Application 2006/0159339. Sek M. Chai, Mohamed Ahmed, Nikos Bellas, Greg Kujawa, King F. Lee, Abelardo Lopez Lagunas. "Method and apparatus as pertains to captured image statistics".
- [P2] US Patent Application 2006/0262140. Greg Kujawa, Mohamed Ahmed, Nikos Bellas, Sek M. Chai, King F. Lee, Abelardo Lopez Lagunas. "Method and apparatus to facilitate visual augmentation of perceived reality".
- [P1] US Patent Application 2008/0120497. Sek M. Chai, Nikos Bellas, Malcolm Dwyer, Dan Linzmeier. Automated configuration of a processing system using decoupled memory access and computation

INVITED TALKS, PANELS, TUTORIALS

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| 09.2025 | MLSysOps project. Invited talk at the Workshop of AI Computing Continuum: Beyond Now. European Health and Digital Executive Agency (HaDeA). Brussels, Belgium. |
| 01.2024 | Reconfigurable Architectures for Approximate Dense SLAM Computing in Robotics. Invited talk at the STEADINESS workshop in conjunction with HiPEAC 2024. Munich, Germany. |
| 04.2023 | The present and future of computing: Domain-Specific Accelerator. Invited Talk at the 14th Electrical and Computer Engineering Student Conference (ECESCON). Volos, Greece. |
| 07.2017 | Reliability and Energy-efficiency using Significance-Based Computing. Invited Talk at Design Test Verification and EDA Workshop. Volos, Greece. |
| 02.2017 | Reliability and Energy-efficiency optimizations using Significance-Based Computing. Invited Talk at Stanford Research Institute (SRI), Princeton, NJ. |
| 10.2016 | Reconfigurable Computing. Invited Talk at the University of Thessaly IEEE Student Branch. |
| 09.2016 | Approximate computing for next-generation platforms. Talk ARM Research Summit. Cambridge, UK. |
| 04.2016 | A Benchmark Suite at the Meeting Point of Heterogeneous and Approximate Computing. Invited Talk at the Workshop on Benchmarking and |

Measuring Approximate Computing (BMAC) held in conjunction with ISPASS 2016. Uppsala, Sweden.

- 03.2016** Reliability and Energy-efficiency optimizations using Significance-Based Computing. Keynote Talk in Workshop on Parallel Programming for Resilience and Energy Efficiency (PP4REE). Barcelona, Spain.
- 03.2015** Enablers and Roadblocks for Mainstream Adoption of Approximate Computing Paradigm. Panel at WACAS 2015 in conjunction with ASPLOS 2015. Istanbul, Turkey.
- 10.2014** Significance-Based Computing for Reliability and Power Optimization. Invited Talk at HiPEAC Computing Systems Week. Athens, Greece.
- 09.2014** Significance-Based Computing for Reliability and Power Optimization. Invited Talk at the 6th International Conference on Numerical Analysis. Chania, Greece.
- 06.2014** Significance-Based Computing for Reliability and Power Optimization. Invited Talk at the Department of Informatics and Telecommunications, University of Athens, Greece.
- 05.2014** Significance-Based Computing for Reliability and Power Optimization. Talk at Aristotle University of Thessaloniki, GR.
- 01.2014** SCoRPiO: Significance Based Computing for Reliability and Power Optimization. Invited Poster at the Workshop on Energy Efficient Electronics and Applications (WEEE). Lausanne, Switzerland.
- 05.2012** Massively Parallel Programming Models Used as Hardware Description Languages: The OpenCL Case. Invited Talk at University of Toronto, Toronto, ON.
- 03.2012** Massively Parallel Programming Models Used as Hardware Description Languages: The OpenCL Case. Invited Talk at Aristotle University of Thessaloniki, GR.
- 10.2011** Massively Parallel Programming Models Used as Hardware Description Languages: The OpenCL Case. Invited Talk at Ecole Polytechnique Fédérale de Lausanne (EPFL).
- 10.2011** Reliable computing on faulty substrates: a preliminary study. Invited Talk at Ecole Polytechnique Fédérale de Lausanne (EPFL).
- 08.2011** Massively Parallel Programming Models Used as Hardware Description Languages: The OpenCL Case. Invited Talk at Stanford Research Institute (SRI), Princeton, NJ.
- 03.2007** Automatic Generation of Streaming Accelerators from a High Level Representation. Invited Talk at the Department of Electrical Engineering, University of California at Riverside, CA.
- 09.2006** Programming models and Architectures for Reconfigurable Platforms. In-

- vited Talk at Motorola Software, Systems, Simulation (S3) Symposium, Itasca, IL.
- 03.2006** Programming Models and Architectures for FPGAs. Motorola Symposium on Innovations in DSP and Embedded Systems Design using FPGAs, Schaumburg, IL.
- 09.2005** Automatic Generation of Imaging Architectures from an Algorithmic Representation. Motorola Software, Systems, Simulation (S3) Symposium, Itasca, IL.
- 06.2003** Imaging Technologies at Motorola Labs. Invited Talk at the Computer Engineering and Informatics Department (CEID), University of Patras, Greece.
- 02.2000** Loop-Cache: An Instruction Hierarchy Component for Reduced Energy Consumption. Invited Talk at the Department of Electrical and Computer Engineering, Northwestern University, Evanston, IL.
- 09.1999** Architectural and Compiler Techniques for Energy Reduction in High Performance Processors. Invited Talk at the Department of Electrical and Computer Engineering, Purdue University, W. Lafayette, IN.
- 12.1998** Architectural and Compiler Techniques for Energy Reduction in High Performance Processors. Talk at IBM Research, Austin, TX.
- 11.1998** Architectural and Compiler Techniques for Energy Reduction in High Performance Processors. Talk at Microprocessor Research Labs, Intel Corp. Portland, OR.
- 08.1996** Novel techniques for Power reduction in High Performance Processors. Design Technology Labs, Intel Corp., Santa Clara, CA.

AWARDS AND HONORS

- 05.2021** Best paper award. ACM International Conference on Computing Frontiers (CF).
- 02.2021** Best paper award nominee. Design, Automation & Test in Europe Conference & Exhibition (DATE).
- 06.2014** Best paper award nominee. IEEE International Conference on Application-specific Systems, Architectures and Processors (ASAP).
- 06.2000** IEEE Transactions on VLSI. Special issue on the most significant advances in techniques and methodologies for power-conscious design.
- 11.1992** First Rank, Commencement ceremony of School of Engineering, University of Patras, Greece.
- Greek Scholarship Foundation academic excellence awards for being in the top 5 in the Department of Computer Engineering and Informatics.

SERVICE

Committee and Editorial Services

- **General Chair/Organizer:** ML4ECS (26, 25), WAPCO (17, 16, 15), ECVW (09).
- **Program Chair:** ECVW (08).
- **Technical Program Committee:** ICPP(24), SMACD(24), CCGrid(20), DATE(24, 22, 21, 20), DAC(19, 18, 17), ARC(26, 25, 19, 18), EUC(13, 12), FPL(11), MICRO(07), ECVW(09, 08, 07).
- **Guest Editor:** Journal on Computer Vision and Image Understanding. Special issue on embedded computer vision (10).
- **Other Roles:** Local Arrangements Chair MICRO(21), Publications Chair MICRO(07), Workshops Chair FPL(11).

Reviewer

ACM TODAES, ACM TACO, ACM TRTS, ACM TOMCCAP, IEEE Micro, IEEE TCOMP, IEEE TVLSI, IEEE TCAD, IEEE TETC, IEEE TSUSC, IEEE TPDS, Springer JSPS, Elsevier SUSCOM, Elsevier Parallel Computing, Elsevier JPDC, MICRO, FPL, CCGrid, Supercomputing, ICPP, ICS, ISLPED, ISCA, DAC, DATE, ECVW, ARC, HPDC, ISC, ICPP, UbiComp, BMVC, SRDS, WAPCO, Motorola Labs Patent Committee.

Professional Associations

- IEEE Member
- Member of the European Network of Excellence on High Performance and Embedded Architecture and Compilation (HiPEAC)
- Member of Technical Chamber of Greece

Departmental Services

- Director of the Graduate MSc Program of ECE Department at UTH (2018-2020, 2023-)
- Member of the Graduate MSc committee of ECE Department at UTH (2014-2020)
- Member of the Graduate PhD committee of ECE Department at UTH (2016-2020)
- Member of the Undergraduate Curriculum Committee of ECE Department at UTH (2019)

PHD SUPERVISION

Primary supervisor

- Alexandros Patras - ECE Department, University of Thessaly. Dissertation area: Machine Learning techniques to improve execution flow. In progress.
- Maria-Rafaela Gkeka - ECE Department, University of Thessaly. Dissertation area: Efficient and Reliable Visual SLAM Architectures. In progress.
- Konstantinos Parasyris - ECE Department, University of Thessaly. Dissertation area: System software techniques to enhance reliability of modern platforms. (PhD 10.2018). Now Computer Scientist at Lawrence Livermore National Labs (LLNL), CA, USA.

- Muhsen Owaida - ECE Department, University of Thessaly. Dissertation area: Architectural Synthesis using Parallel Programming Models (PhD 09.2012). Now Senior DSP Engineer at Magic Leap, Zurich, Switzerland.

Member of PhD committee

- Alexandros Koumbaroulis - ECE Department, University of Thessaly. Dissertation area: Audio Visual Speech Processing in the Wild. In progress.
- Panagiotis Koutsovasilis - ECE Department, University of Thessaly. Dissertation area: Virtual Machine Policies for Improving Energy Efficiency. (PhD 03.2020).
- Christos Kalogirou - ECE Department, University of Thessaly. Dissertation area: Software Policies and Mechanisms for Improving Performance and Lowering Power Consumption for the Memory Subsystem of High Performance Computers. (PhD 03.2020).
- Emmanuel Koutsoumbelias - ECE Department, University of Thessaly. Dissertation area: Programming and Testing Support for Drone Based Applications (PhD 09.2018).
- Vassilis Vassiliadis - ECE Department, University of Thessaly. Dissertation area: Optimization of Program Execution Using Computational Significance (PhD 11.2017).
- George Georgakoudis - ECE Department, University of Thessaly. Dissertation area: Scheduling and Performance Characterization on Heterogeneous Computing Systems (PhD 05.2016).
- Nikolaos A. Foutris Department of Informatics and Telecommunications, University of Athens. Dissertation area: Architectures for Dependable Modern Microprocessors (PhD 02.2016).
- Antonios S. Nikitakis Department of Electronic and Computer Engineering, Technical University of Crete. Dissertation area: High Performance Low Power Embedded Vision Systems (PhD 10.2013).
- Georgios Kornaros Department of Electronic and Computer Engineering, Technical University of Crete. Dissertation area: Real Time ASIC Monitoring for System-Level Power and Thermal Management (PhD 09.2013).
- Ioannis Mavroidis Department of Electronic and Computer Engineering, Technical University of Crete. Dissertation area: Algorithm Mapping to Reconfigurable Systems and Systems with Multiple Embedded Processors (PhD 12.2011).
- Dimitrios Syrivelis - ECE Department, University of Thessaly. Dissertation area: Exploiting Reconfigurable Heterogeneous Parallel Architectures in a Multitasking Context: A systems Approach (PhD 06.2009).
- Dimitrios Bountas - ECE Department, University of Thessaly. Dissertation area: EDA tools for Power and Reliability Estimation of ICs (PhD 06.2009).
- Dimitrios Karampatzakis - ECE Department, University of Thessaly. Dissertation area: Reliable ICs (PhD 06.2009).
- Somsubhra Mondal - ECE Department, Northwestern University. Dissertation area: Architectural Optimizations and CAD Tools for Improved Energy Efficiency and Faster Design Closure for FPGAs (PhD 06.2007).

MSC SUPERVISION

- Maria Rafaela Gkeka - ECE Department, University of Thessaly. Dissertation area: Simultaneous Localization and Mapping (SLAM) using the Vulkan Programming Model. In progress.
- Konstantinos Parasyris - ECE Department, University of Thessaly. Dissertation area: System software techniques to enhance reliability of modern platforms (MSc July 2017).
- George Delis - ECE Department, University of Thessaly. Dissertation area: Implementation of a Mobile Wireless Sensor Node for the Replacement of Failed Nodes (MSc March 2014).
- Konstantis Daloukas - ECE Department, University of Thessaly. Dissertation area: Compiler and Run-Time Support for OpenCL on Hardware- and Software-Managed Cache Multicores (MSc July 2010).

RESEARCH FUNDING

- **Project Title:** Machine Learning for Autonomic System Operation in the Heterogeneous Edge-Cloud Continuum (MLSysOps).
Role: co-investigator. One of the authors of the proposal.
Source of funding: AI-enabled computing continuum from Cloud to Edge. HORIZON-CL4-2022-DATA-01-02.
Budget and dates: Total project budget 5.7 million . Budget share 692,000 . 1/2023-12/2025.
- **Project Title:** GPU/FPGA programming Educational seminar.
Role: co-investigator.
Source of funding: StartTech Ventures EURL.
Budget and dates: 10,000 . Budget share 10,000 . 11/2021 2/2022.
- **Project Title:** Very Low Power GPUs for Mobile Robotics and Virtual Reality (vipGPU).
Role: Scientific Coordinator. Main author of the proposal.
Source of funding: Greek General Secretariat for Research and Technology, Competitiveness, Entrepreneurship and Innovation Operational Programme.
Budget and dates: Total project budget 1 million . Budget share 260,000 . 2018-2021.
- **Project Title:** A Universal Micro-Server Ecosystem by Exceeding the Energy and Performance Scaling Boundaries (UniServer).
Role: co-investigator. One of the authors of the proposal.
Source of funding: Low power computing, H2020-ICT-2015.
Budget and dates: Total project budget 4.8 million . Budget share 480,250 . 2/2016-2/2019.
- **Project Title:** Significance-Based Computing for Reliability and Power Optimization (SCoRPiO).
Role: Coordinator. Main author of the proposal.
Source of funding: FET-Open Programme, FP7-ICT-2011-C.
Budget and dates: Total budget 1.9 million . Budget share 420,000 . 6/2013-6/2016.
- **Project Title:** System Software for Future, Heterogeneous, Accelerator-Based Systems (Centaurus).

Role: co-investigator. One of the authors of the proposal.

Source of funding: Greek Ministry of Education, Lifelong Learning and Religious Affairs, Aristeia II Program.

Budget and dates: Total budget 250,000 , 7/2014-11/2015.

- **Project Title:** Advanced Mathematical Methods and Software Platform for solving Multiphysics, Multi-Domain Problems on Modern Computer Architectures: Application to Environmental Engineering and Medical Problems (MATENVMED).

Role: co-investigator. One of the authors of the proposal.

Source of funding: Greek Ministry of Education, Lifelong Learning and Religious Affairs, Thales Program, grant ID: 137.

Budget and dates: Total budget 600,000 , 7/ 2012-11/2015.

- **Project Title:** Automatic Hardware Generation Using the Streaming Paradigm.

Role: Coordinator. Main author of the proposal.

Source of funding: Marie Curie International Reintegration Grant (IRG), FP7.

Budget and dates: 100,000 (11/2008-11/2012).

- **Project Title:** Low Power Microprocessor Design.

Role: Coordinator. Main author of the proposal.

Source of funding: Intel Corp.

Budget and dates: \$45,000 per year for 3 years (1996-1998).

TEACHING

Course instruction

- **Computer Organization and Design** (ECE219). University of Thessaly. 2nd year class. (Fall 25 - Fall 13).
- **Introduction to Computer Systems** (ECE134). University of Thessaly. 1st year class. (Spring 13 - Spring 09).
- **Parallel Computer Architecture** (ECE340). University of Thessaly. 4th year class. (Spring 26 - Spring 13, Fall 11 - Fall 08).
- **Embedded Systems** (ECE338). University of Thessaly. 3rd year class. (Spring 26 - Spring 08).
- **Advanced Topics in Computer Architecture** (ECE658). University of Thessaly. Graduate class. (Fall 25 - Fall 07).

Curriculum development

- **Computer Organization and Design** (ECE219): Major revision.
- **Parallel Computer Architecture** (ECE340): Developed from the beginning.
- **Embedded Systems** (ECE318): Developed from the beginning. Emphasis on FPGA-based MPSoCs.
- **Advanced Topics in Computer Architecture** (ECE658): Developed from the beginning.
- **Advanced Topics in Computer Architecture** (Northwestern University): Developed from the beginning. Graduate course in collaboration with professor Gokhan Memik at Northwestern University.